Phase 3 Project Design Phase

Student Enrollment System • Academic Year 2026–2027

Powered by Salesforce

PROBLEM SOLUTION FIT

Problem-Solution Fit: Bridging the Gap

CURRENT: Manual Work

Manual Work

Spreadsheets & Paper Forms

Data Loss

Inconsistent records & errors

Fragmented Data Across Departments

Silos between departments • No shared visibility

VS

SOLUTION: Centralized with Salesforce

Centralized Salesforce

Unified data platform • Single source of truth

Automated Processes

Automated workflows • Reduced manual effort

Real-time Visibility

Live dashboards • Instant reporting & insights

PROPOSED SOLUTION — 4 MAIN MODULES: DEEP DIVE

Student Management

- **Records & Profiles** centralized student profiles
- **Admissions** application tracking & onhoarding
- **Academic History** grades, progress, transcripts

Course Management

- **Catalog & Curriculum** — courses, programs, details
- **Scheduling** • classes, timetable, instructor assignments
- **Prereguisites & Evaluation** • prerequisite logic, grading & assessments

KEY BENEFITS

Consistent, accurate, reliable data

Automation

Streamlined processes, time savings

Scalability

Easily scales to institutional needs

Phase 3 — Project Design Phase

Student Enrollment & Course Management System

Academic Year 2026–2027

Problem–Solution Fit

Problem

- Manual work: spreadsheets & paper forms
- Data loss: inconsistent records & errors
- Fragmented data across departments

Solution

- Centralized Salesforce system
- Unified data and automated processes
- Real-time visibility & reliability

Proposed Solution — 4 Main Modules

1. Student Management

- Student records & profiles
- Admissions tracking
- Academic history

2. Course Management

- Course catalog & curriculum
- Scheduling & prerequisites
- Course evaluation

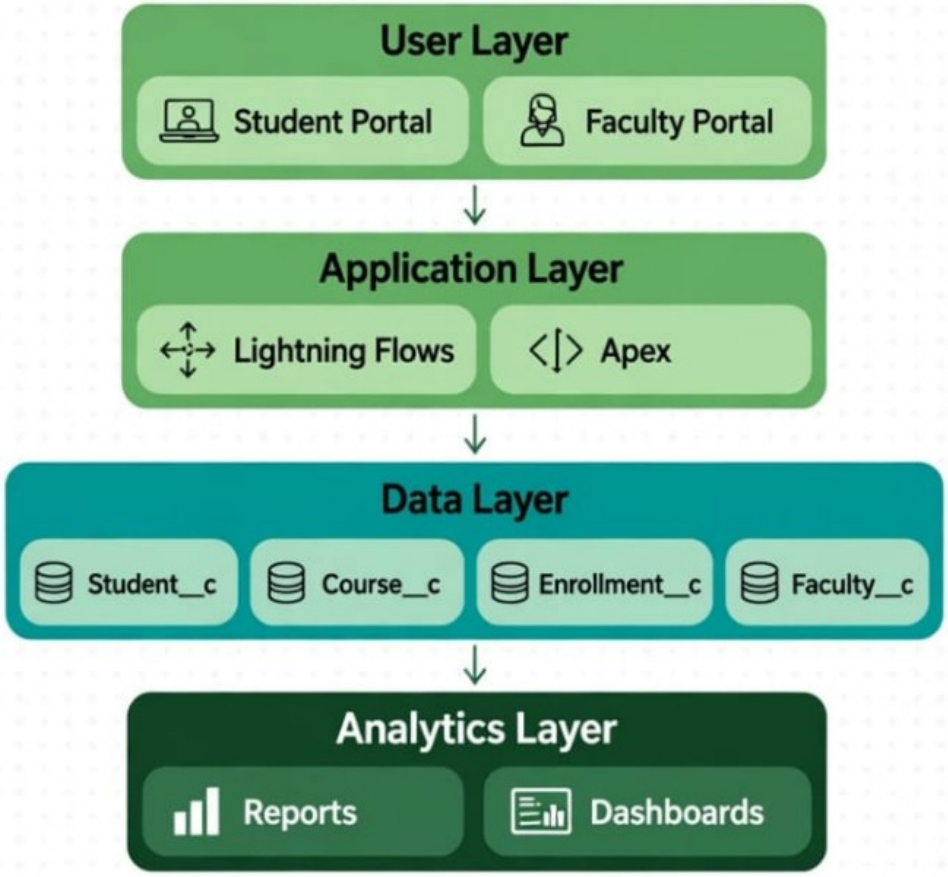
3.Enrollment Management

- Registration workflow
- Waitlist & approval process
- Enrollment status tracking

4. Faculty Management

- Faculty profiles & assignments
- Workload management
- Performance tracking

Solution Architecture



Key Benefits: Data Integrity • Automation • Scalability • Role-Based Access • Real-Time Insights

Student Enrollment System

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Enrollment Management — Deep Dive

Registration Workflow

1. Student Submits Registration → 2. Eligibility Validation → 3. Seat Availability Check → 4. Enrollment Confirmed

Waitlist Approval Process

Step 1: Waitlisted when capacity is full

Step 2: Admin Review & Criteria: registation date
- academic priority
- program requirement

Step 3: Approved → Enrolled
Enrolled → Denied → Notified Student

Enrollment Status Tracking

Pending | Approved | Waitlist | Dropped

Real-time status updates via automated notifications & alerts

Faculty Management

Faculty Profiles
- Credentials & Qualifications
- Department & Contact
- Expertise & Specialization

Assignments
- Course Assignments
- Section Capacity Management
- Schedule Alignment

Workload
- Credit Hours Tracking
- Class Count per Semester
- Hours/Week Monitored

Light: 6-9 hrs | Standard: 10-12 hrs | Heavy: 13+ hrs

Performance Tracking
- Student Feedback Scores
- Evaluation Ratings
- KPI Metrics
- Tenure Review

Solution Architecture — 4 Layer Design

Layer 1: User Layer

Student Portal
- Registration & Course Browse
- Enroll/Drop Courses
- View Enrollment
- Notifications & Updates

Faculty Portal
- View Class Roster
- Enter Grades
- Manage Assignments
- Approve Waitlist

Layer 2: Application Layer

- Lightning Flows
- Apex Triggers
- LWC (Lightning Web Components)
- Validation Rules — Data Integrity & Input Validation

Layer 3: Data Layer

- Student Records & Profile
- Course Catalog & Details
- Enrollment__c
- Faculty Records & Department

Relationship: Enrollment__c serves as many-to-many junction between Student__c and Course__c. Faculty__c is linked to Course__c via one-to-many relationship.

Layer 4: Analytics Layer

Data Model — ER Diagram Visualization

Student__c
- Student_ID (PK)
- Name
- Email
- Program
- Status

(1) → (N)

Enrollment__c (Junction)
- Enrollment_ID (PK)
- Student_ID (FK)
- Course_ID (FK)
- Status: Pending | Approved | Waitlist | Dropped
- Enrollment_Date
- Enrolled_Date

Course__c
- Course_ID (PK)
- Course_Name
- Credits
- Capacity
- Semester

(M) — (M)

Faculty__c
- Faculty_ID (PK)
- Faculty_Name
- Department
- Workload_Hrs
- Performance_Rating

Faculty__c

Key Benefits

- **Improved Efficiency** — Automated registration reduces manual processing by ~60%
- **Real-Time Visibility** — Live dashboards for enrollment & faculty workload monitoring
- **Scalable & Secure** — Built on Salesforce platform with role-based access & security
- **Data-Driven Decisions** — KPIs enable proactive capacity planning & resource allocation
- **Enhanced User Experience** — Self-service portals empower students & faculty
- **Reduced Administrative Overhead** — Automated workflows & approvals reduce manual effort